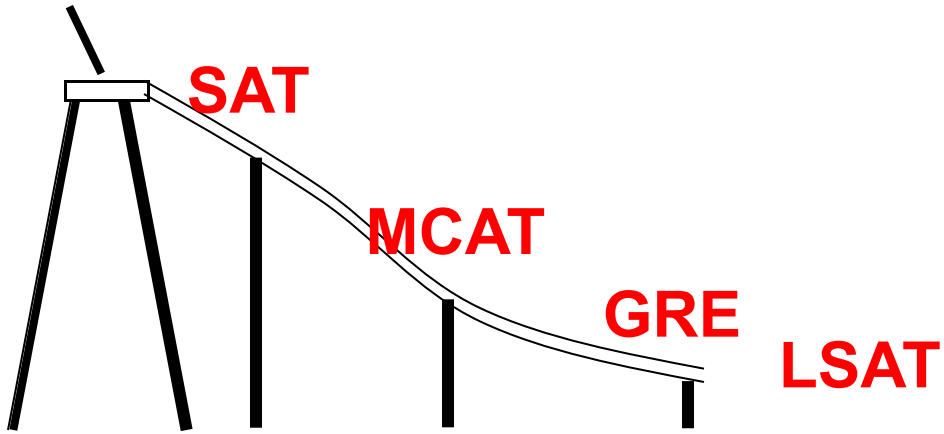
Test Slide



Quantum Computing



Dave Bacon

CSE 326 Special Lecture

Final Logistics

- Section A: (Bacon) 8:30 - 10:20 MGH 231
- Section B: (Anderson) 10:30 - 12:20 MGH 241

- Solution Sets, Example Final, Study guide available on website

- Office hours:

  Ruth : Wed March 14, 9-5 or so
  Dave : Should be around much of the week

A Challenge



188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

A Game



Alice

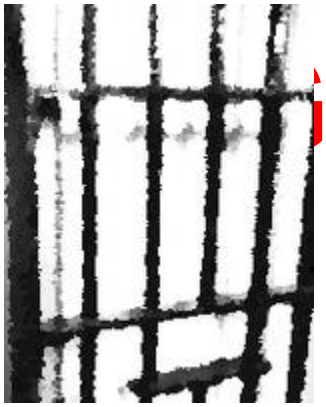


Bob



Charlie

An Evil Game







Alice **Bob** **Charlie**

1. Each given slip **X** or **Y**

2. At midnight each party shouts "+1" or "-1"

Winning the Game







Alice	Bobb	Charlie	Winning Product
X	X	X	+
X	Y	Y	1
Y	Y	X	- 1
Y	X	Y	- 1

A Game







Alice

A

X

A

Y

Bob

B

X

B

Y

Charlie

C

X

C

Y

All either +1 or -1

A Game







Alice	Bob	Charlie	Product
X	X	X	AXBXCX
X	Y	Y	AXBYCY
Y	Y	X	AYBYCX
Y	X	Y	AYBXCY

Winners?

Assuming winning strategy:

AXBXCX = +1

AXBYCY = +1

AYBYCX = +1

AYBXCY = +1

Multiply equations….contradiction!

LHS:

AX2AY2Bx2BY2Cx2CY2=+1

Losers



Alice



Bob



Charlie

There is no strategy to win the game

In the Real World







Alice

Bob

Charlie

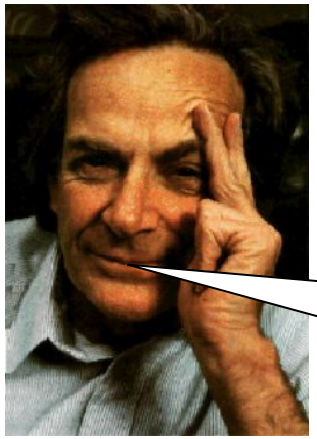
This is wrong, there is a way to win

Quantum Theory

Welcome to the Wonderfully Wacky World Of Quantum Theory

Please Check Your Notion of Reality at the Door

Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize (1965)

Anyone who is not shocked by quantum theory has not understood it.

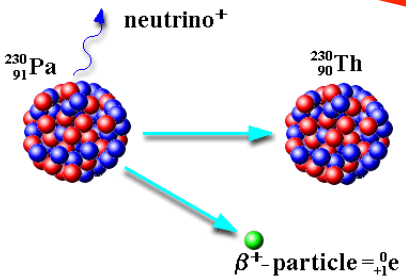


Niels Bohr
Nobel Prize (1922)

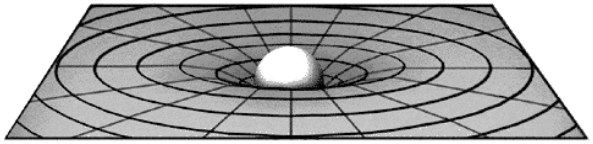

hogwash...

What is Quantum Theory?

Weak Force



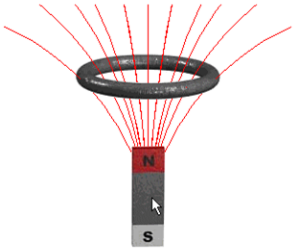
Gravitational Force (?!?!)



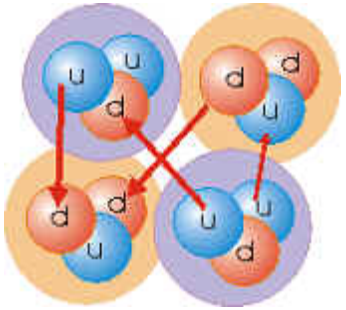


Quantum Theory

Electromagnetic Force



Strong Force



Physic's "computer language"

Where is Quantum Theolry?

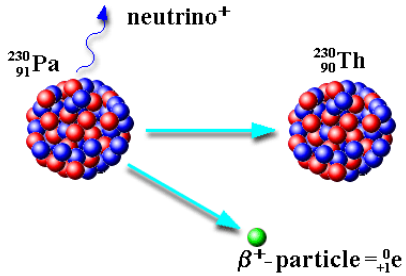
Quantum

Classical

small

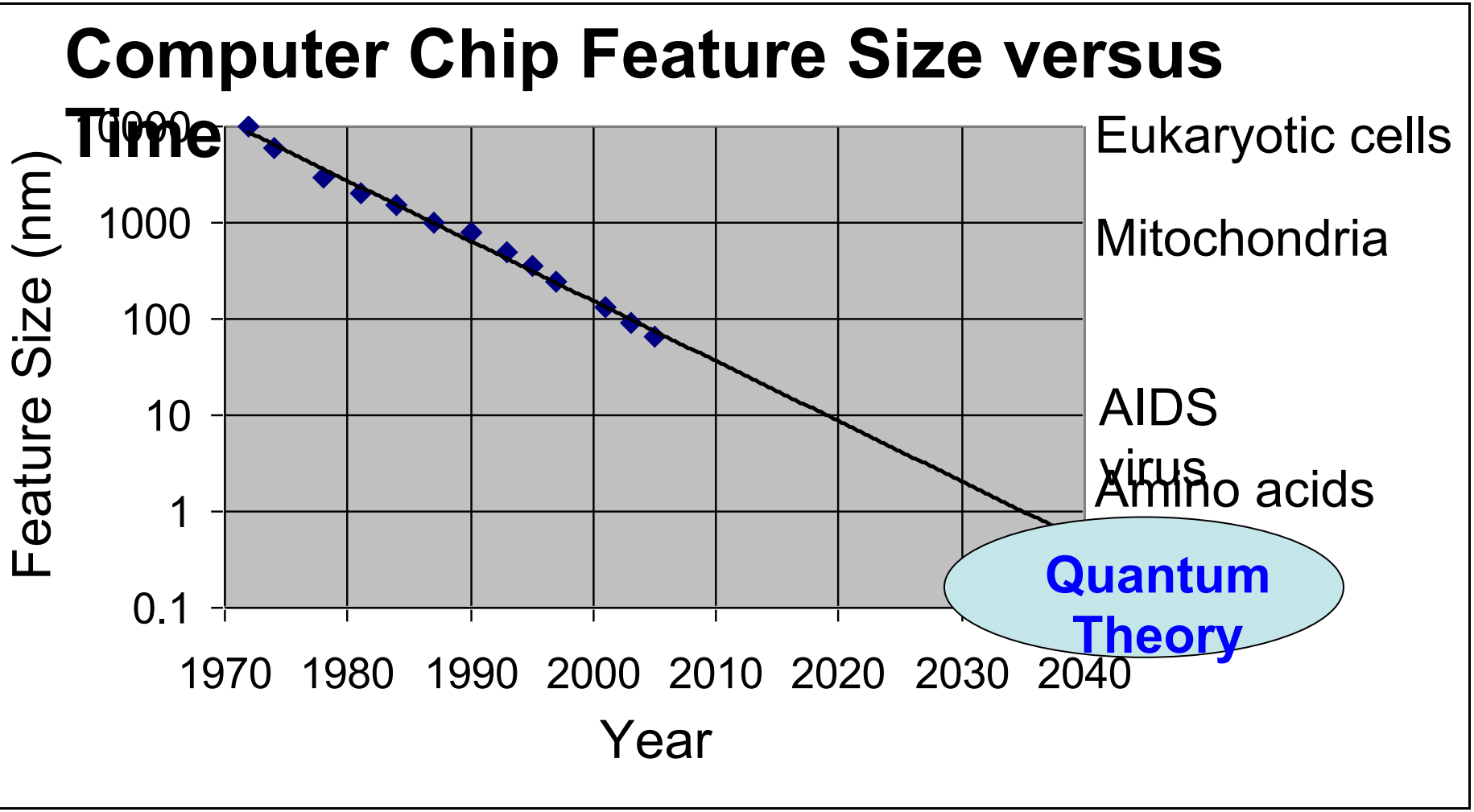
roughly size

large





Moore's Law



"Prediction is very difficult, especially if it's about the future."

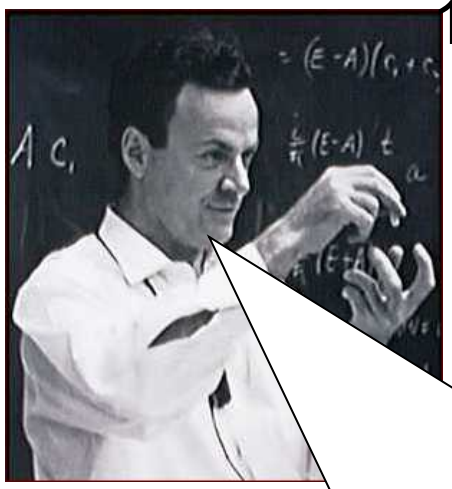


Niels Bohr

A Worrying Physicist

2040 is a long time in the future….so why worry?

Richard Feynman



Observations

1. No efficient computer simulations of quantum theory.
2. Systems evolving according to quantum theory efficiently compute properties of the quantum system (that's what experiments are!)

"How can we simulate quantum mechanics?... Can we do it with a new kind of computer…a **quantum computer**? It's not a Turing machine but a machine of a different kind" (1981)

Quantum Theory Made Easy

Classical

probabilities

0	p_0
1	p_1

pi is a real number

$p_0+p_1=1$

$Prob(i)=p_i$

bit

Quantum

amplitudes

0	a_0
1	a_1

ai is a complex number

$|a_0|^2+|a_1|^2=1$

$Prob(i)=|a_i|^2$

qubit

Superpositio

Qubits

	amplitudes
0	a_0
1	a_1

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle$$

ai is a complex number

Schrödinger's Cat

$$\frac{1}{\sqrt{2}}\left(|\ \right.$$



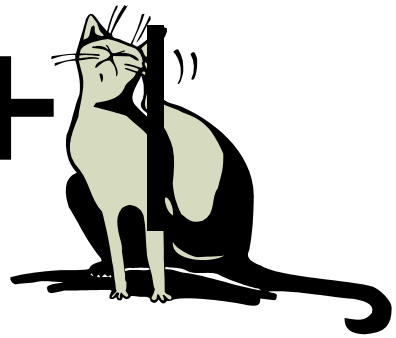
$$\left.\ \rangle + |\ \right.$$



$$\left.\ \rangle\right)$$

Commercial Philosophy Break

Dead or Alive?

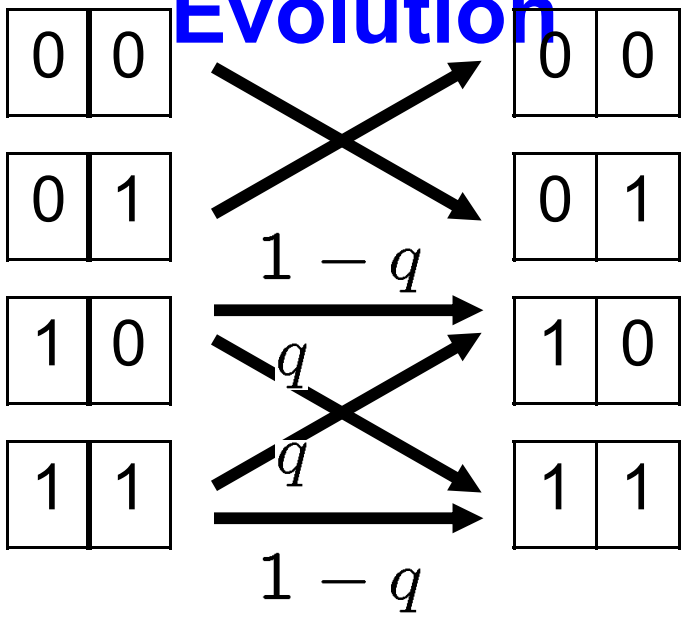
$$\frac{1}{\sqrt{2}}\left(\left|\right.\right.$$  $$\left.\left.\right\rangle + \left|\right.\right.$$  $$\left.\left.\right\rangle\right)$$

Dr. Thurman: Do you feel alone right now?

Donnie Darko: I don't know. I mean I'd like to believe I'm not, but I just… I've just never seen any proof so I… just don't debate it any more it's like I could spend my whole life debating it over and over again, weighing the pros and cons, and in the end I still wouldn't have any proof. So I just, I just don't debate it any more. <laughs> It's absurd. The search for God is absurd.

Quantum Theory Made Easy

Classical Evolution

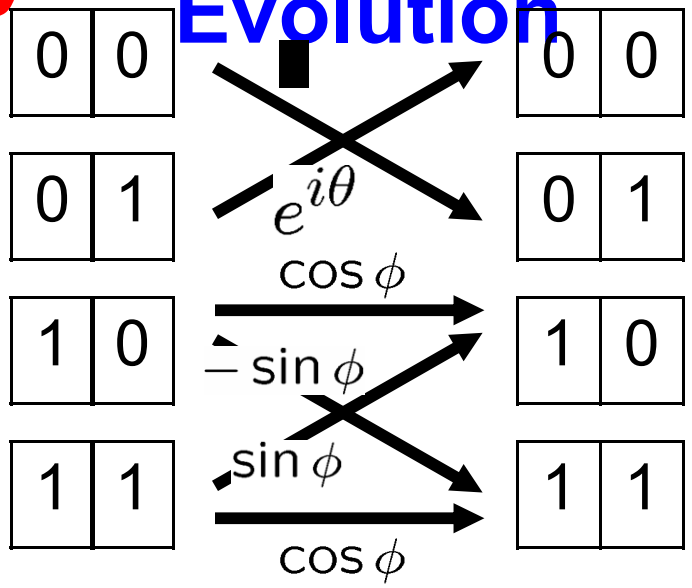


transition probabilities

$$\begin{bmatrix} p'_0 \\ p'_1 \\ p'_2 \\ p'_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1-p & q \\ 0 & 0 & p & 1-q \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

stochastic matrix

Quantum Evolution



transition amplitudes

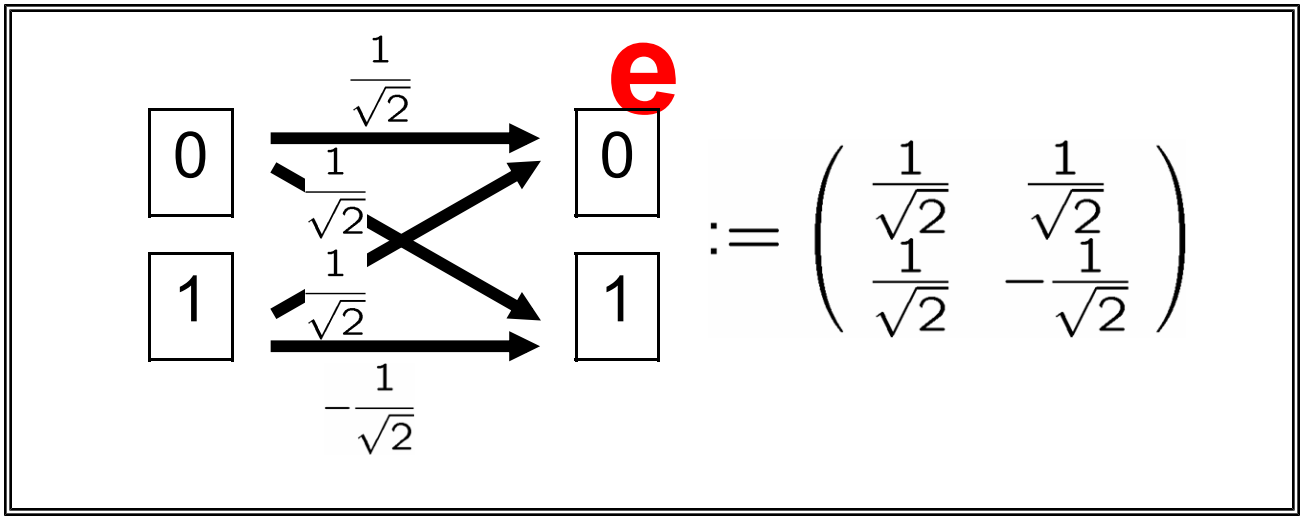
$$\begin{bmatrix} a'_0 \\ a'_1 \\ a'_2 \\ a'_3 \end{bmatrix} = \begin{bmatrix} 0 & e^{i\theta} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & \cos\phi & \sin\phi \\ 0 & 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

unitary matrix



Interferenc e





qubit input

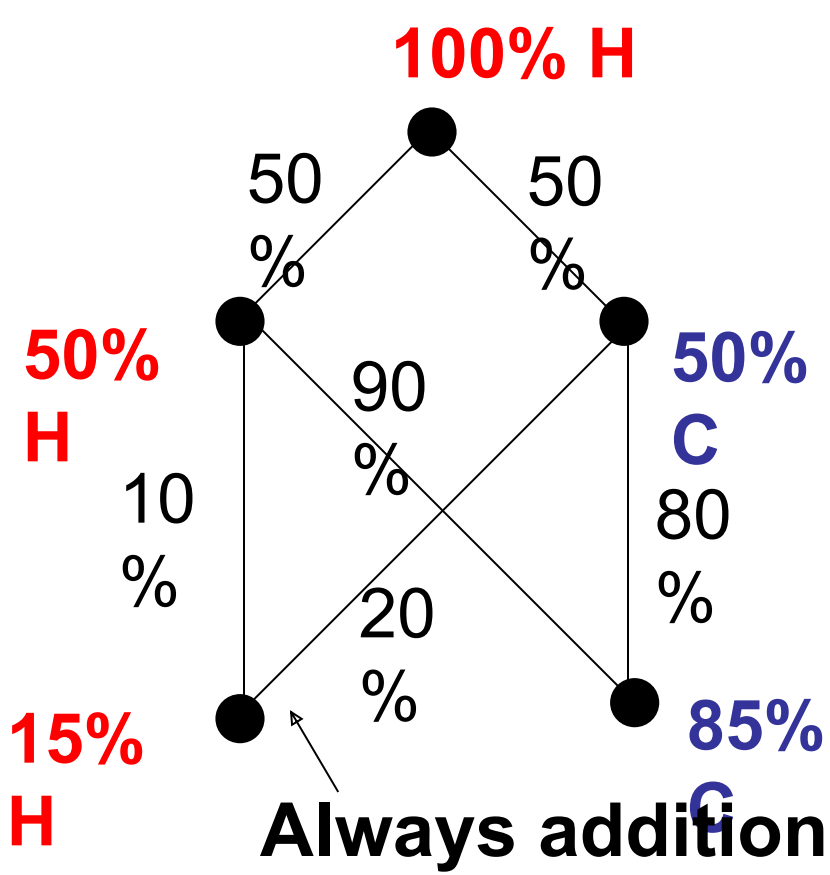
$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

0	50 %	100%
1	50 %	0%

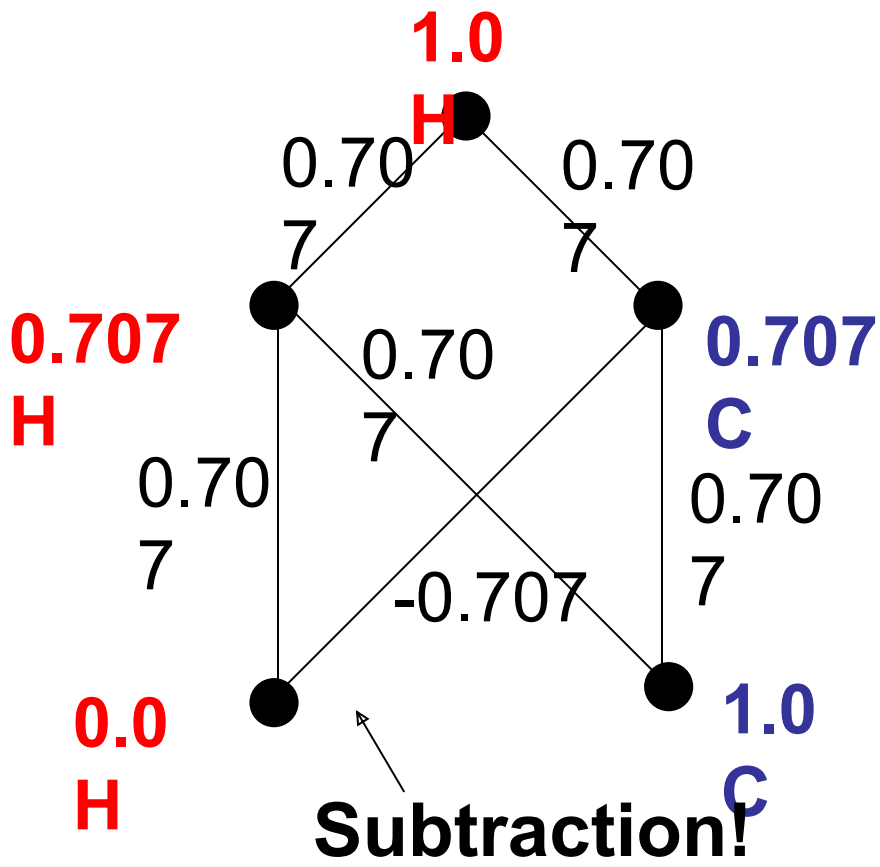
$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

0	50 %	0%
1	50 %	100%

Interfering Pathways

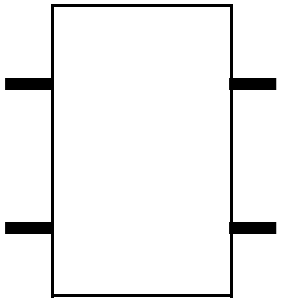


Classical



Quantum

Deutsch's Problem



output



David Deutsch



Dr. Falcon

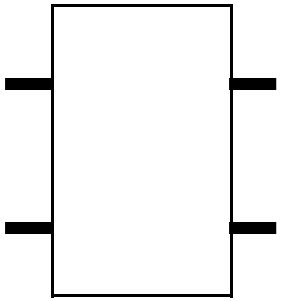
$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	constant		balanced

Deutsch's Problem:

How many times do we have to use this black box to determine whether we are given the first two or the second two?

Deutsch's Problem

$$\frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$



$00 \rightarrow 00$	$00 \rightarrow 01$	$00 \rightarrow 00$	$00 \rightarrow 01$
$01 \rightarrow 01$	$01 \rightarrow 00$	$01 \rightarrow 01$	$01 \rightarrow 00$
$10 \rightarrow 10$	$10 \rightarrow 11$	$10 \rightarrow 11$	$10 \rightarrow 10$
$11 \rightarrow 11$	$11 \rightarrow 10$	$11 \rightarrow 10$	$11 \rightarrow 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
constant		balanced	

$$\frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle \qquad \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

$$-\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle \qquad -\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$

Deutsch's Problem

$$
\begin{array}{c} \\ 00 \\ 01 \\ 10 \\ 11 \end{array}
\begin{array}{c}
\begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\
\begin{bmatrix}
\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\
\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\
\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\
\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2}
\end{bmatrix}
\end{array}
$$

$$\frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle \rightarrow |01\rangle$$

$$-\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle \rightarrow -|01\rangle$$

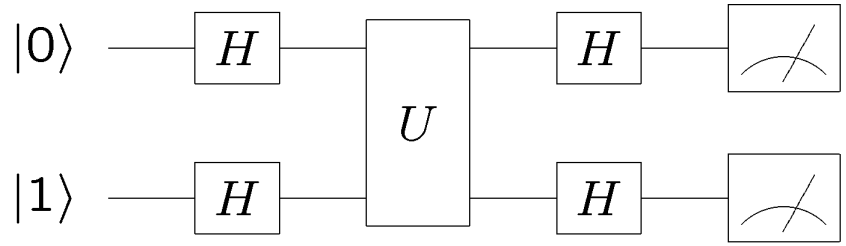
constant

$$\frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle \rightarrow |11\rangle$$

$$-\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle \rightarrow -|11\rangle$$

balanced

Quantum Deutsch



Classica **2 Uses**

Quantum **1 Use**



David Deutsch

1st Quantum Speedup

1985

Computational Complexity

How many basic computational steps does a computation take?

Multiplication

$$\begin{array}{r} 152 \\ \times\ 421 \\ \hline 152 \\ 304 \\ +\ 1008 \\ \hline 103992 \end{array}$$

Input: 2 n digit numbers

Number of operations: about n2

Scales as polynomial in input size

A Challenge



188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

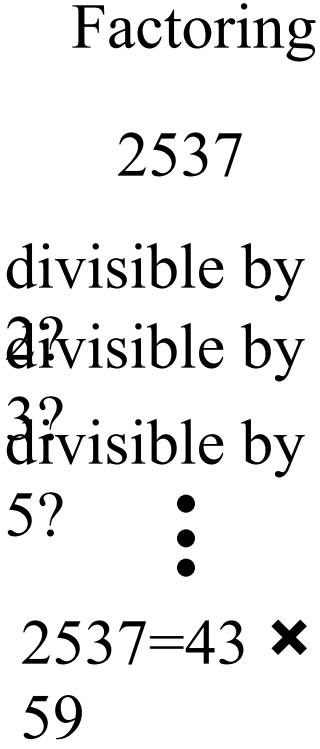
pq

$$
\begin{aligned}
&3980750864240649373971255005503864911990643623425267\\
&0840638518957594638895726176858331\\
&\times\\
&4727721461074353025362230719730482246329146953020971\\
&1645985217113052071125636359039752 7
\end{aligned}
$$

Computational Complexity

How many basic computational steps does a computation take?

Factoring



Input: n digit numbers

Number of operations: about 10n/2

Scales as exponential in input size

One Time Pads

Random n bit string

0010101111010001



Alice



Bob

Alice's message

0110110011100101

secret key

$\oplus$

0100011100110100

secret key

$\oplus$

0110110011100101

Eve cannot learn message

Public Key Cryptography

Interesting history:

1st schemes “known in public” where put forth by
Diffie and Hellman in 1976 (key exchange) and
Rivest, Shamir and Adleman in 1978 (encryption algorithm)
(based on work by Merkle in 1974, published 1978)

However, it now appears that the British researchers working for British intelligence (GCHQ) were actually the first to discover these protocols, but their work was classified at the time!
Cooks in 1973 (encryption algorithm)
Williamson in ~1973 (key exchange)
(based on work by James Ellis in the late 1960s.)

Public Key Cryptography Roughly



Alic

Instructions for how to make her lock.



Bob

Bob's secret documents





Assume: very hard to design key from instructions to make lock

This is (very roughly) what happens in public key cryptography

Public Key Encryption RSA



Alic
e



Bob

1. Alice generates two random large primes p q and $N = pq, \phi = (p-1)(q-1)$
2. Alice chooses a number $1 < e < \phi$ which is coprime with ϕ.
3. Alice computes d such that $de = 1 \mod \phi$

Public Key: (N, e)

Private Key: (N, d)

Public Key Encryption: RSA



Public Key: (N, e)



Alic
e

Bob

Public Key: (N, e)

Private Key: (N, d)

Bob's message: n

$$c = n^e \bmod N$$

$$c = n^e \bmod N$$

$$c^d \bmod N = n^{ed} \bmod N$$

$$= n \bmod p = n \bmod q \quad \text{(FLT)}$$

$$= n \bmod pq = n \bmod N \quad \text{(CRT)}$$

Public Key Encryption: RSA



Alic
e

Public Key: (N, e)

Private Key: (N, d)



Bob

Bob's message: n

Bob, using public key can encrypt message.

Alice, using private key, can decrypt the message

But decrypting without the private key is (thought) to be computationally hard

Public Key Encryption: RSA



Alic
e

Public Key: (N, e)

Private Key: (N, d)



Bob

Bob's message: n

<u>Factoring can be used to break RSA</u>

If we could factor, $N = pq$ then we could compute $\phi = (p-1)(q-1)$ from which you could use e to find d

$$de = 1 \mod \phi$$

Then we just use the standard decryption:

$$c^d \mod N = n \mod N$$

Shocking scoveries



Umesh Vazirani (U.C. Berkeley)

Quantum computers offer superpolynomial speedups (for certain oracle promise problems) over classical computers [Bernstein, Vazirani 1993]

Quantum computers offer exponential speedups (for certain oracle promise problems) over classical computers [Simon 1994]



Dan Simon (Microsoft)

Shor's Algorithm



Peter Shor 1994

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

39807508642406493739712550055038649119906436234252670840638518957594638895726176858331794727721461074353025362230719730482246329146953020971164598521711305207112563635903975271

×

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^3 \log n)$

Shor's Algorithm

Breaks RSA Cryptography!



"Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world's most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security. "

-Simon Singh

Order-Finding and Factoring

Factor *N*

choose *x* coprime to *N* (Euclid's algorithm for gcd)

Order finding: find smallest *r* such that $x^r = 1 \mod N$

Use order finding to factor: suppose r is even,

$x^r - 1 = 0 \mod N$ N divides $x^r - 1$

N divides $(x^{\frac{r}{2}} - 1)(x^{\frac{r}{2}} + 1)$

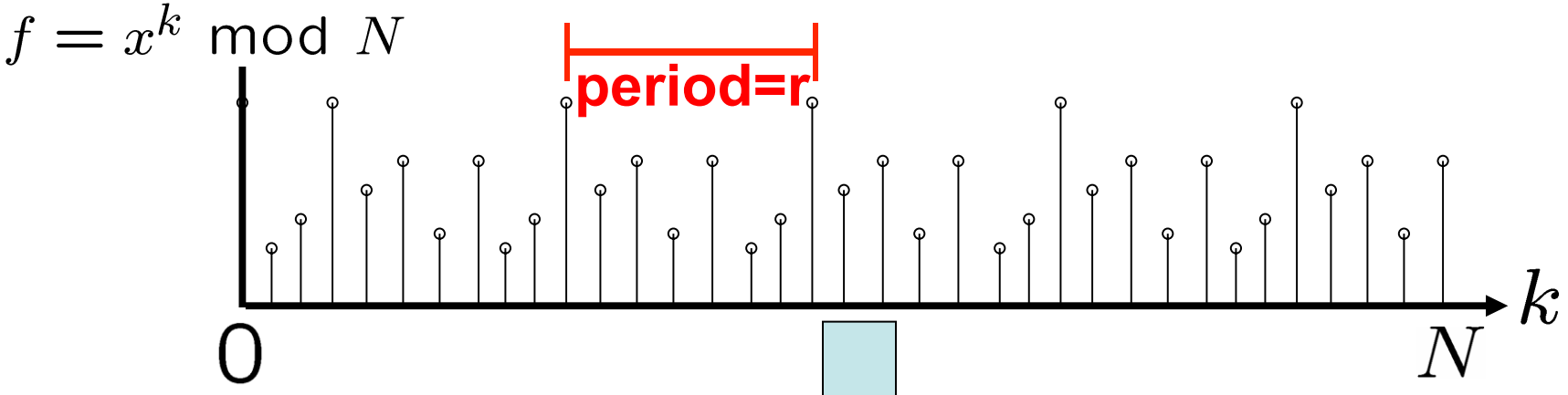
But $1 \leq x^{\frac{r}{2}} \leq N$ N must share a common factor N with $x^{\frac{r}{2}} \pm 1$ not equal to

If *r* is even then compute $\gcd(x^{r/2} \pm 1, N)$ as factor!

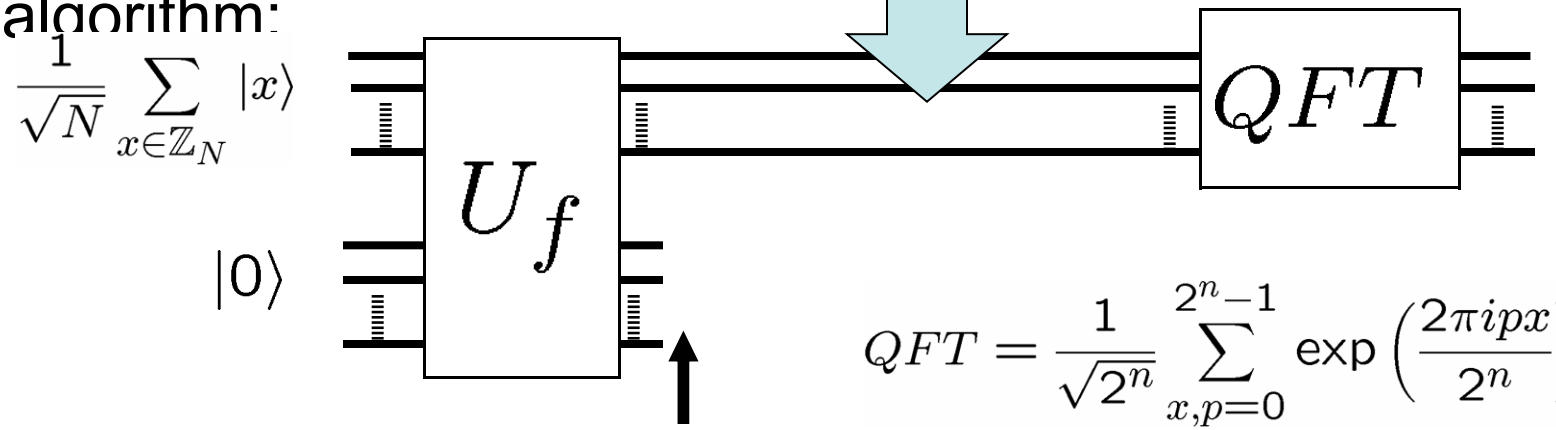
More tricky: r is even happens with high probability

Shor's Algorithm

Factoring can be reduced to period finding.



Anatomy of Shor's algorithm:



Measure and use to determine r

$$QFT = \frac{1}{\sqrt{2^n}} \sum_{x,p=0}^{2^n-1} \exp\left(\frac{2\pi i p x}{2^n}\right) |p\rangle\langle x|$$

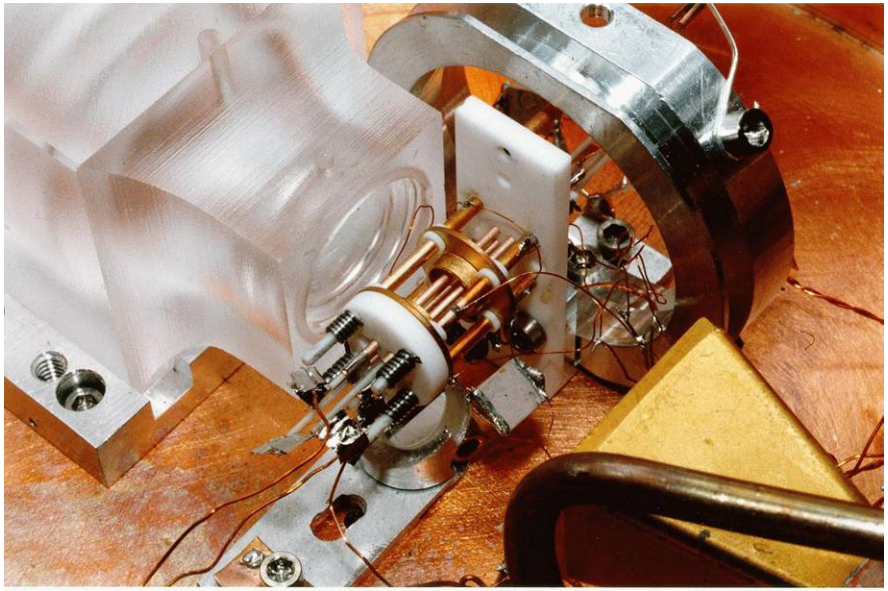
$$\frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} |x\rangle \otimes |f(x)\rangle$$

Quantum Computers

Nature is Quantum!

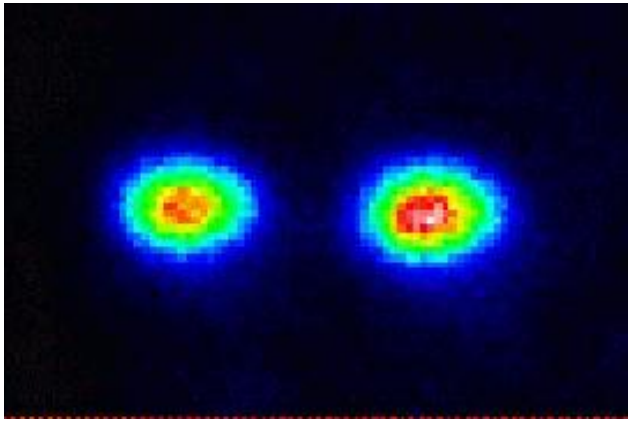
The most fundamental theory of computation must be Quantum!

Ion Trap Quantum Computers

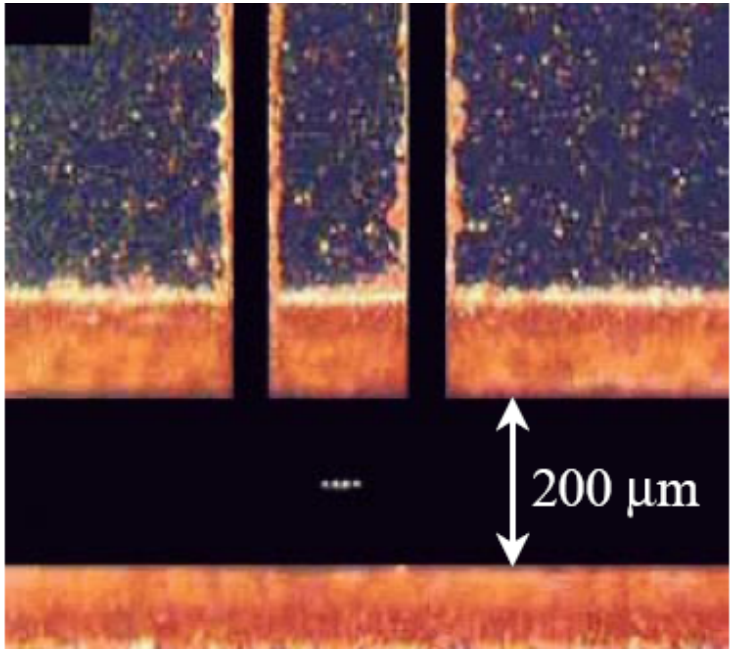


Oscillating electric fields trap ions

like charges repel



2 9Be+ Ions in an Ion Trap



In the Real World







Alice

Bob

Charlie

This is wrong, there is a way to win

Quantum Theory

GHZ Sta



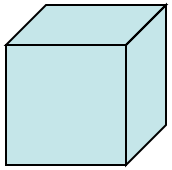


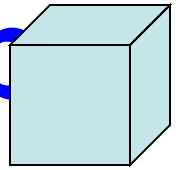


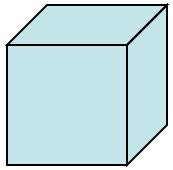
Alic e

Bo b

Charli e







$$\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle$$

GHZ Prot







Alic e

Bo b

Charli e

Each party applies:

$$M_X := \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$M_Y := \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

GHZ Magic







Alic **Bo** **Charli**

XX $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow \frac{1}{2}|000\rangle+\frac{1}{2}|011\rangle+\frac{1}{2}|101\rangle+\frac{1}{2}|110\rangle$

XX $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow -\frac{1}{2}|111\rangle+\frac{i}{2}|001\rangle+\frac{1}{2}|100\rangle+\frac{i}{2}|010\rangle$

XY $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow -\frac{1}{2}|111\rangle+\frac{i}{2}|001\rangle+\frac{i}{2}|100\rangle+\frac{1}{2}|010\rangle$

XX $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow -\frac{1}{2}|111\rangle+\frac{1}{2}|001\rangle+\frac{i}{2}|100\rangle+\frac{i}{2}|010\rangle$

Y $0 \rightarrow +1 \quad 1 \rightarrow -1$

In the Real World







Alice

Bob

Charlie

By sharing “entangled” quantum particles, Alice, Bob, and Charlie win!

Quantum Information Science

Coming age of quantum devices

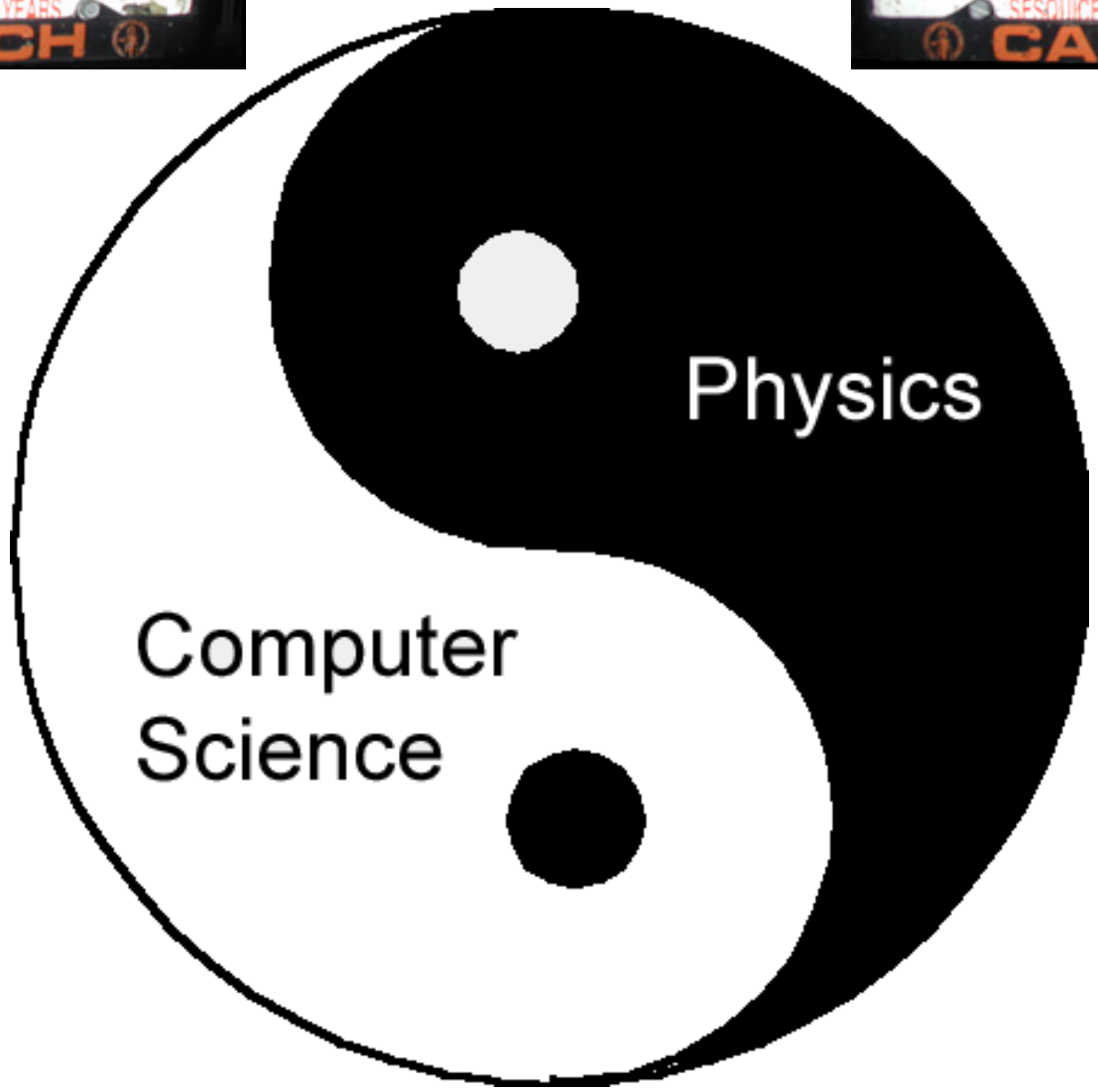
Foundations should be based on quantum theory

Ultimate physical limits to computation?


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